



Abu Sayem Md Siam

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PROFESSIONAL SUMMARY

About me

I am an Associate Software Engineer with hands-on experience in developing scalable microservice web applications, SaaS platforms, and data-driven solutions. At RoBenDevs, I've contributed to building microservice-based CRM systems, designing robust backends, and integrating third-party tools to enhance business performance. I enjoy tackling complex problems, from optimizing ETL pipelines to automating workflows that streamline operations. I focus on writing clean, maintainable code while collaborating with cross-functional teams to deliver reliable, high-quality software. Passionate about learning and growth, I am eager to keep expanding my expertise in data science to deepen my knowledge in machine learning, statistical modeling, and big data technologies, ultimately contributing to building innovative, user-focused data solutions.

SKILLS

Programming with Python, Node js, Ruby, Bash, React, Js | Machine Learning | Python | Git | SQL | Numpy | Scikit-Learn | Matplotlib | Data Science | Jupyter Notebook | Data Analysis | Data Visualization | Anaconda | Seaborn | Neural Networks and Deep Learning | Artificial Intelligence | Keras | TensorFlow | Computer Vision | Convolutional Neural Networks | PyTorch | Data Preprocessing | Deep Learning | Github | Object-Oriented Programming

WORK EXPERIENCE

SOFTWARE ENGINEER – ROBENDEV'S – 1 Jul 2024 – Current – DHAKA, BANGLADESH

Business or Sector: Information and communication | **Department:** Software

- **Developed Custom CRM Systems:** Architected and implemented a scalable Customer Relationship Management (CRM) system based on microservice architecture, enabling modular functionality, high availability, and seamless integration with third-party services.
- **Built and Maintained SaaS Products:** Engineered Software-as-a-Service (SaaS) solutions, focusing on robust backend development, API design, and user-centric features to ensure scalability, performance, and customer satisfaction.
- **Developed Data-Intensive Applications:** Created and optimized data-intensive applications, including Extract, Transform, Load (ETL) pipelines, to process large volumes of data efficiently, ensuring data integrity and performance.
- **Automated Business Processes:** Designed and deployed automation workflows to streamline business operations, reducing manual effort and improving operational efficiency through custom scripts and tools.
- **Integrated Marketing Tools:** Implemented and managed integrations with marketing tools such as Google Tag Manager (GTM) and tracking pixels, enabling accurate analytics, campaign tracking, and enhanced marketing performance.
- **Ensured System Reliability and Performance:** Monitored and optimized application performance, conducted regular maintenance, and implemented best practices for security, scalability, and fault tolerance.

- **Collaborated with Cross-Functional Teams:** Worked closely with product managers, designers, and stakeholders to gather requirements, define technical specifications, and deliver high-quality software solutions on time.
- **Maintained Code Quality and Documentation:** Followed best practices for code quality, including unit testing, code reviews, and comprehensive documentation to ensure maintainability and knowledge transfer.

● **EDUCATION AND TRAINING**

19 JAN 2020 – 1 MAY 2024 Dhaka, Bangladesh
BACHELOR OF SCIENCE (BSC) IN COMPUTER SCIENCE AND ENGINEERING United International University (UIU)

- Acquired comprehensive knowledge in programming languages including C, C++, Python, JavaScript, Java, and PHP, enabling the development of robust and scalable software solutions.
- Studied Digital Logic Design, gaining expertise in designing and optimizing digital circuits and systems.
- Developed proficiency in Computer Networks and Computer Security, focusing on network protocols, cybersecurity principles, and secure system design.
- Mastered Machine Learning and Deep Learning, applying algorithms and models to solve complex data-driven problems.
- Gained expertise in Software Engineering, Database Management, System Design, and Software Architecture Patterns, with hands-on experience in designing scalable, maintainable, and efficient systems.
- Conducted research in the fields of Data Science, resulting in 4 papers published in international conferences and 1 paper published in a peer-reviewed journal.

Website <https://www.uiu.ac.bd/> | **Field of study** Information and Communication Technologies |

Final grade CGPA- 3.74 Out of 4.00 | **Level in EQF** EQF level 6 | **National classification** NQF Level 7 |

Type of credits Credit Hour System | **Number of credits** 137 |

Thesis Decentralized Medical Image Sharing: A Blockchain Based Approach with Subject Sensitive Hashing for Enhanced Privacy and Integrity

● **LANGUAGE SKILLS**

Mother tongue(s): **BENGALI**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C1	B2	B2	B2	B2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

● **PUBLICATIONS**

2025
[FVCM-Net: Interpretable privacy-preserved attention driven lung cancer detection from CT scan images with explainable HiRes-CAM attribution map and ensemble learning](#)

This paper introduces FVCM-Net, a novel framework merging federated learning, attention mechanisms, and ensemble learning for privacy-preserving lung cancer detection from CT scans. Federated learning enables collaborative training across institutions without data sharing, enhancing model generalizability. Ensemble methods boost prediction accuracy, while XAI tools like SHAP and HiResCAM provide interpretability. Using datasets from LIDC-IDRI, IQ-OTH/ NCCD, Kaggle, and others, FVCM-Net attained 98.26% accuracy and 97.37% F1-score, aiding radiologists in clinical decisions and advancing medical imaging.

Authors: Abu Sayem Md Siam, Md Mehedi Hasan, Yeasir Arafat, Md Muzadded Chowdhury, Sayed Hossain Jobayer, Fahim Hafiz, Riasat Azim | **Journal Name:** Journal Biomedical Signal Processing and Control | **Volume, Issue and Pages:** Volume 112, Pages 108719 | **Publisher:** Elsevier

2025
[Decentralized Medical Image Sharing: A Blockchain Based Approach with Subject Sensitive Hashing for Enhanced Privacy and Integrity](#)

This research introduces a blockchain-based framework for secure medical image sharing, emphasizing data integrity and privacy. It comprises two phases: JPEG compression with feature extraction via deep neural network-based subject-sensitive hashing (SSH) to maintain diagnostic quality, and encryption with storage on IPFS. SSH ensures consistent feature maps between original and compressed images, while encrypted data and hashes enable secure retrieval through smart contracts that validate access requests. Experimental validation across multiple datasets yields a 98% average correctness rate, addressing challenges in scalable, private healthcare data management and advancing reliable solutions for medical imaging in clinical systems.

Authors: Yeasir Arafat, Abu Sayem Md Siam, Md Muzadded Chowdhury, Md Mehedi Hasan, Sayed Hossain Jobayer, Swakkhar Shatabda, Salekul Islam, Saddam Mukta | **Journal Name:** IET Blockchain | **Volume, Issue and Pages:** Vol. 5 Issue 1, Pages e70009 | **Publisher:** John Wiley & Sons Inc

2024

Deep Learning for Identification of Skin Diseases: Emphasizing on Arsenic-Induced Skin Conditions

Skin diseases like arsenic-induced lesions, melanoma, psoriasis, and eczema pose diagnostic challenges due to visual similarities, with accurate classification vital for arsenic-related conditions. This study develops a deep learning model, ARS-CNNNSA, for identifying arsenic-related and common skin diseases using a custom dataset. Evaluating pre-trained models such as VGG16 (85% accuracy), MobileNetV2, InceptionV3 (82%), and ConvNeXt Base, ARS-CNNNSA excelled with 91% accuracy and 90% F1-score. Integrating Conv2D layers and self-attention mechanisms, it captures complex features while minimizing over-fitting and hyper-parameters. This approach enhances diagnostic precision in arsenic-prone areas and lays groundwork for AI tools in dermatology.

Authors: Sayed Hossain Jobayer, Md Muzadded Chowdhury, Abu Sayem Md Siam, Umma Hani Mim, Md Mazedul Hasan Khan Zidan, Mansora Akter Bithe, Md Mehedi Hasan | **Journal Name:** 2024 27th International Conference on Computer and Information Technology (ICCIT) | **Volume, Issue and Pages:** Pages-3254-3259 | **Publisher:** IEEE

2024

SE-VGG16 MaizeNet: Maize Disease Classification Using Deep Learning and Squeeze and Excitation Attention Networks

Deep learning has transformed plant disease identification, enhancing accuracy for rapid diagnosis in agriculture. This study analyzes maize leaf disease detection using deep transfer learning models on Plant Village and PlantDoc datasets. Evaluated models include VGG16, ResNet50, and others, with a novel integration of attention mechanisms into VGG16 and ResNet50 for improved feature focus. The VGG16+SE composite achieved 93.44% validation accuracy, while MobileNetV2 led with 94.76%. These advancements enable early detection, aiding farmers and researchers in preventing outbreaks, boosting crop yields, and reducing agricultural losses through efficient, AI-driven tools for maize health management.

Authors: Abu Sayem Md Siam, Abir Hossain, Rifat Bin Hossain, Md Musfiqur Rahman | **Journal Name:** 2024 International Conference on Emerging Smart Computing and Informatics (ESCI) | **Volume, Issue and Pages:** Page 1-6 | **Publisher:** IEEE

2023

TextileNet: A Deep Learning Approach for Textile Fabric Material Identification from OCT and Macro Images

This study introduces a novel deep learning and transfer learning approach for textile fabric classification, addressing limitations in traditional methods' accuracy and efficiency. Utilizing two datasets—Optical Coherence Tomography (OCT) images for detailed structural insights and macro images for diverse samples—we evaluated pre-trained CNN architectures. Deep learning automatically extracts key fabric features from raw data, while transfer learning enhances performance with minimal training data. Comparative analysis highlights MobileNetV2's superior results, achieving 99.87% accuracy on OCT images and 95% on macro images. These findings demonstrate MobileNetV2's potential for real-world fabric classification in the textile industry, improving automation and precision.

Authors: Abu Sayem Md Siam, Yeasir Arafat, Md Mushfikur Talukdar, Md Mehedi Hasan, Raiyan Rahman | **Journal Name:** 2023 26th International Conference on Computer and Information Technology (ICCIT) | **Volume, Issue and Pages:** pages 1-6 | **Publisher:** IEEE

2023

Bangla News Classification Employing Deep Learning

In this paper, we have introduced a deep learning model for the classification of Bangla news articles as it is an important task for maintaining and managing Bangla news articles. The proposed model combines a hybrid Recurrent Neural Network (RNN) comprising Bi-directional Long-Short Term Memory (LSTM) and Bi-directional Gated Recurrent Unit (GRU) algorithms to improve classification accuracy. We tested the performance of proposed classifier with traditional machine learning techniques e.g. naïve Bayes and decision tree induction. The experimental results indicate that the deep learning models achieved almost 90% accuracy. This study provides valuable insights into the

performance of various machine learning and deep learning techniques for Bangla news classification and highlights the potential of deep learning for this task.

Authors: Abu Sayem Md Siam, Md Mehedi Hasan, Md Mushfikur Talukdar, Md Yeasir Arafat, Sayed Hossain Jobayer, Dewan Md Farid | **Journal Name:** International Conference on Intelligent Systems and Data Science | **Volume, Issue and Pages:** 155-169 | **Publisher:** Springer Nature Singapore

● PROJECTS

VentureVest: A Startup and Investor Platform.

Developed using Django, SQLite, HTML, CSS, Ajax and Bootstrap. Enables startups to showcase products, raise funds, and share monthly updates. Investor can communicate through in built-in chat.

Link <https://github.com/Duranta19/VentureVest>

UIU Helpline: Student Support and Knowledge Sharing.

Developed using PHP, MySQL, HTML, CSS, and Bootstrap. Uploading academic books and questions of regular evaluations. Community (Discussion Forum) and problem-solving section

Link <https://github.com/Duranta19/UIU-Helpline>

Roots: Product Supply Chain Management System.

Developed using PHP, MySQL, HTML, CSS, Bootstrap and Google Charts. A supply chain management system builds a connection between the farmer, distributor and retailer. Monitoring the current market and product price.

Link <https://github.com/Duranta19/Root>

Alumni-Linked: Connects Students and Alumni.

Developed using PHP, MySQL, HTML, CSS, Ajax and Bootstrap. Build connections between students and alumni. View alumni profiles, job postings, the Discussion Forum, Company Reviews, apply for jobs, and job preparation.

Link https://github.com/Duranta19/ALUMNI_LINK

● ACHIEVEMENTS

Academic Scholarship

- 100% Scholarship in 3 trimesters for excellent academic performance in UIU
- 50% Scholarship in 3 trimesters for excellent academic performance in UIU
- 25% Scholarship in 1 trimester for excellent academic performance in UIU